The Law of Graphic Equalisation — Physics Grounding

Change, c, the Reconciliation of Special and General Relativity, the Dimensional Constants, and the Second Law

Matthew Parslow
Independent Researcher

Draft — draft-2026-08-26
26 August 2026

DRAFT — draft-2026-08-26 (26 August 2026). This document is under active revision and is circulated for comment, not as a finished result. It is a companion to *The Law of Graphic Equalisation — Theory* (referred to below as Graphic Equalisation), and inherits that paper's definitions, epistemic boundary, and level-of-claims. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

This companion to Graphic Equalisation develops the physics groundings of the change-substrate. Its central move is that a frame's time is an inferred count of *cycles* and that the invariant speed c is the frame's own change-throughput unit, 1 by construction rather than by postulate; from that one economy special and general relativity, the dimensional constants, and the second law follow as *shapes*. Special relativity's c-invariance becomes a theorem; special and general relativity reconcile as one fixed change-budget spent two ways, on motion or on change-load; the dimensional constants $(c, \hbar, k_{\mathrm{B}}, G)$ are each a frame's own base unit and their magnitudes are cross-frame artefacts; the uncertainty principle is the finite-aperture floor, the Landauer bound the information–energy conversion, quantization the aperture cell, and the Planck scale the frontier. The second law is no-copy accumulated, with entropy frame-relative, the demon exorcised by cost, and local order affordable as paid minting. The claims are information-level, physics taken as an embodiment: the paper mechanises the shapes and explicitly does not derive the field equations, the dimensionless constants, or a theory of quantum gravity.

1 Scope

This paper is a grounding, not a physical theory. It inherits Graphic Equalisation's position that the claims are about information — distinction, frame, admissibility, realisation — of which physical theory is one embodiment, and that a physical result bears on an information-level claim when it settles an informational fact. Everything below is at that level. The strong claims are structural: that there is a c, a $\hbar$, an uncertainty principle, a Landauer floor, a second law, and that special and general relativity reconcile as one change-economy. The weak part is anything carrying a number: no field equations, no quantitative metric, no exact Lorentz factor, no dimensionless constant, and no theory of quantum gravity. Each grounding is a *promoted* constraint exposed to falsification, in the manner of Graphic Equalisation's collapse bridge, not a result asserted from within.

2 c as the change-to-time rate

Graphic Equalisation establishes that time is not a substrate dimension but is *inferred* by a frame, whose unit is a *cycle* — one traversal of a closed rule-loop, of no fixed length. Once time is the inferred count of cycles, the invariant speed c has a natural reading, consistent with the missing-denominator result (there is no rate *in* substrate-time, since there is no substrate-time to divide by). c is the fixed *rate* at which change is realised — per unit of the substrate's throughput, a cycle consuming a variable amount of it — from which a frame's time is inferred. So c is prior to time: the conversion between change-count and inferred duration, which is exactly what a constant relating the temporal and spatial parts of an interval is.

Proposition 2.1 (c is each frame's own unit, hence invariant by construction)**.** *c is each frame's own base rate, its change-throughput, so every frame measures its own c as the unit,* 1 *— and a unit cannot be measured as anything else. Its invariance is therefore a theorem, not a postulate: not that all frames happen to agree on a speed, but that each is reporting its own unit.*

A value *other* than 1 is a cross-frame quantity: the familiar number appears only when one frame is compared with another — change-throughput against a spatial metric, seconds against metres — which is where the continuous lives, in the maps between frames rather than in a frame's own discrete states. One can give time a number at all only by comparing frames; in its own frame c is the unit, and its particular value (299 792 458 in SI) is an artefact of the comparison, not a fact about the substrate. And c is a *limit* because a signal is a sequence of changes and none can be realised faster than the frame's throughput.

3 The change-budget: speed, dilation, mass, aperture

Speed is a quantum of that unit. The fixed throughput c splits between *spatial motion* and *internal change* (proper time), and a speed is the fraction spent on motion. At rest the whole budget is internal change and the clock ticks fully; approaching c moves budget to motion, leaving less for internal change, so the clock slows — time dilation as budget re-allocation — and at c the whole budget is motion and no proper time remains. Mass is the budget *reserved* for internal change, so a massive graph can never put all of it into motion: it asymptotes to c and the energy to approach diverges, while a massless one reserves none and *is* at c, with no proper time.

Because observation is funded by internal-change throughput, approaching c coarsens a frame's *aperture*. The transformation between frames in relative motion — time dilation and length contraction — is an aperture transformation between different budget-splits, and each frame's aperture self-calibrates so that its own c still reads 1. The Lorentz factor is the aperture change that holds $c = 1$ across the split. (The constant-magnitude four-velocity of Minkowski geometry (Minkowski, 1909) is this budget-split stated as a metric fact rather than as a mechanism.)

4 Special and general relativity reconciled

The two relativities are one mechanism, because the one fixed budget c can be drained two ways.

Proposition 4.1 (SR and GR are one change-budget spent two ways)**.** *Special relativity is the budget spent on* motion*: a moving graph puts throughput into spatial motion, leaving less for internal change, so its clock slows. General relativity is the budget spent on* change-load*: mass is throughput tied up in internal change, and concentrating change-load (dense mass) consumes the local budget, so local clocks slow. Both dilations are the same act — the one fixed budget drained, by motion or by mass, each leaving less for proper time.*

The equivalence principle is the seam. Accelerating (changing one's motion) and standing in a gravitational field (change-load) draw on the budget identically, so inertial mass (resistance to a change of motion — more state to re-write, hence more change and more cost, the impulse relation of the Narrative-momentum paper with mass as the exchange rate) and gravitational mass (change-load consuming throughput and slowing local time) are one quantity, "how much change this matter loads onto the ruleset," measured two ways. General relativity assumes their equality and does not explain it; here they coincide by construction.

Corollary 4.2 (Relativity reconciles because c is defined, not postulated)**.** *Because c is here* defined *(Proposition 2.1) rather than postulated, special relativity's postulate of c-invariance becomes a theorem, and its effects follow as consequences rather than as separate surprises, with the classical paradoxes dissolving into aperture differences — each moving clock right, each reading its own $c = 1$. Special and general relativity then reconcile (Proposition 4.1) as velocity- and gravity-dilation of one budget, joined at the equivalence principle.*

5 Relativity and collapse

Relativity also reconciles with the collapse picture of quantum measurement, by a division Graphic Equalisation already carries: continuous structure lives in the maps between frames, discrete change in a frame's own states. The continuous geometry of general relativity is a cross-frame *map*; the collapse of quantum measurement is the in-frame *event*; and c — the change-to-time rate — is the seam joining the discrete event to the continuous metric. The two are the map and the event of one change-substrate, not rival ontologies, and the standing incompatibility is the artefact of treating them as rivals. This is a claim about the *shape* of the reconciliation, not a derivation: that the field equations follow is not asserted, and no theory of quantum gravity is offered.

6 The measurement problem is frame-relative collapse

Collapse in Graphic Equalisation is a local write that realises one continuation *relative to the frame that interacts.* It is not a physical process needing a mechanism — no dynamical collapse, no spontaneous localisation — but the in-frame realisation that occurs when frames couple. So there is no frame-independent "when": each frame realises its own outcome upon interaction, and a frame that has not interacted holds no collapse. This is Axiom -1 and relational quantum mechanics (Rovelli, 1996) — facts indexed to interactions, not assigned from nowhere.

Proposition 6.1 (Wigner's friend is the map and the event of two frames)**.** *The friend measures the system and realises an outcome — collapse relative to the friend, who interacted. Wigner, not having interacted with friend-and-system, holds relative to himself the continuous unitary* map *(Section 5). Both are correct: the friend holds the in-frame* event*, Wigner the cross-frame* map*, and these are the two sides of the distinction that reconciles general relativity with collapse. The apparent contradiction requires a single frame-independent fact about whether collapse occurred, which Axiom -1 forbids; without that demand the friend's definite result and Wigner's superposition coexist without strain.*

Corollary 6.2 (Any coupling is a measurement; outcomes are definite; no special observer)**.** *Any coupling is a measurement relative to the coupled frames, so no conscious or privileged observer is required and none is distinguished — a plate, a rock, or a friend measure relative to whatever they couple to. Outcomes are definite because collapse realises* one *continuation, and the basis in which they are definite is set by the distinctions the interaction makes — the aperture of the coupling, which is decoherence selecting the pointer basis. There is no cut and no collapse mechanism to locate.*

This dissolves the measurement *problem* — the when, the how, the who, and the definiteness — but it does *not* derive the Born rule. That outcomes are realised frame-relatively is the shape; that their probabilities go as the squared amplitude is a number, and the framework says nothing about it. The same shape/number division applies: the paradoxes are mechanised, the quantitative probabilities handed back untouched.

7 Entanglement is dimensional access through frames

An entangled pair is one system in the superframe, described by a joint dimension — the correlation — that is a first-class quantity of the superframe and NULL in each subframe individually (Graphic Equalisation's higher-frame first-class structure). Each subframe has dimensional access to it only through its own boundary, and only in part; the full dimension lives one level up, along the structural-depth axis rather than the spatial one.

Proposition 7.1 (A subframe's local state is mixed because the purity is a superframe dimension)**.** *Described alone, an entangled particle has no pure state — it is mixed — because its pure description requires the joint dimension, which is* NULL *in its subframe. Tracing out the partner is not discarding locally-available information; the missing information was never a subframe dimension. The reduced state is mixed exactly because the purity is a superframe dimension, inaccessible from below.*

Corollary 7.2 (Nonlocality is up, not across; Bell forbids only subframe-local variables)**.** *The connection between the parties is not* across *the space between them but* up*, to the superframe they are both subframes of. No cross-frame write passes between the subframes; each accesses the same superframe dimension locally, and the joint fact appears only when their accesses are compared (a further coupling), so no signal is sent. Bell forbids* local *— subframe-level — hidden variables (Bell, 1964), and the framework has none: the correlating variable is a superframe dimension, so the violation is expected, not paradoxical. Kochen–Specker contextuality (Kochen and Specker, 1967) is the access itself — measuring projects the superframe dimension into the subframe's aperture, so a realised value is context-dependent by construction.*

Remark (Spookiness is an aperture artefact: the light-switch identity)*.* The confusion entanglement provokes is identical in structure to that of a person who has never seen a light switch and does not understand electricity: flip a switch, a far light obeys, with no connection inside the aperture, so action at a distance is inferred. In both cases the inference is wrong for the same reason — there is a connection, through a dimension the observer cannot see: the wiring in the walls for the switch (the opaque runner), the superframe's joint dimension for entanglement (structural depth). Spookiness is therefore a property of the aperture, not of the phenomenon; a switch is mundane once one sees the wiring, entanglement once one sees the superframe. This is the sequence science has run before — magnetism was action at a distance until the field, gravity until curvature, disease until germs — each "spooky" a real connection through a not-yet-visible dimension, each dissolved by the aperture widening rather than the phenomenon changing. "We do not understand entanglement" is structurally "I do not understand the light switch."

Honest boundary: seeing the connecting dimension dissolves the *spookiness* — the qualitative action-at-a-distance mystery — not the *numbers*. The Tsirelson bound, the correlation strength, and the Born weights are not derived; they become ordinary physics questions rather than spooky ones, the same shape/number division as elsewhere.

8 Amplitudes are a rule on Narrative; the Born rule's frontier

An amplitude is not a foreign number-structure but the directional Narrative of the Narrative-momentum paper, acted on by a rule. That paper's directional momentum $p = \phi_W(\mathcal{N})\,\hat{\mathbf{d}}$ is a

magnitude $\phi(\mathcal{N})$ — the operative-history measure — times a direction $\hat{\mathbf{d}}$, which is a magnitude times a phase: a complex number. The complex amplitude is therefore the directional Narrative, its dynamics a rule applied to it, and its phase the direction $\hat{\mathbf{d}}$ read as position in the cycle (Section 2). The complex structure of quantum amplitudes is thus grounded, not imported.

Proposition 8.1 (The Born square is a candidate via closure)**.** *A realised outcome is a* closed *cycle — the realised "now" is one traversal of the closure loop, and a loop that closes. A probability is the weight of a realised outcome, hence of a closed loop: the Narrative carried forward and carried back to close it, the amplitude and its return. The weight is then the amplitude applied twice, amplitude times its conjugate,* $|amplitude|^2$*. The Born square would on this reading be the closure itself — a realisation must close its loop, and closing requires the directional Narrative both ways.*

This is a candidate, not a derivation, and a real advance on leaving the amplitude a blank. The *complex* structure is grounded (directional Narrative); the *square* has a mechanism (closure); what is not yet done is the rigour — tying closure-as-conjugate to the full Hilbert inner product and Gleason's uniqueness theorem (Gleason, 1957), and showing the phase composition reproduces interference quantitatively. So the Born rule moves from "a structure the framework lacks entirely" to "the directional Narrative for the amplitude, closure for the square, the formal identification still to make." The honest frontier of the physics arc is therefore a *proof* — the rigorous tie to the Hilbert formalism — not a missing object: the framework reaches the amplitude (Narrative), the phase (cycle/direction), and a candidate for the square (closure), and stops at the quantitative formalism, as it stops at every number.

9 Interference and the double slit

The double slit is obvious once two established facts are held together: causality is ordering *in a frame*, and a superframe applies changes to a subframe without passing through the subframe's admissibility.

Proposition 9.1 (Superposition is the possibility structure; interference is amplitudes combining by phase)**.** *Superposition is Graphic Equalisation's possibility structure* $\mathcal{P}(G)$ *— the admissible continuations a graph holds before collapse — so a particle "in superposition" is a graph whose* $\mathcal{P}$ *has more than one admissible continuation, none yet realised. Each continuation carries a directional Narrative (Section 8), an amplitude with a cyclic phase, and before collapse the alternatives' amplitudes combine, constructive where the phases align and destructive where opposed. The outcome weight is the combined closed loop,* $|\sum amplitude|^2$*.*

Two established facts make this un-mysterious. First, causality is ordering *in a frame*: there is no global fact about "which path the particle really took," because which-path is *undetermined*, not *undisclosed*, until a coupling realises it — a cross-frame question with no in-frame answer, like the value of c. The "both paths" is not a particle in two places but the frame's $\mathcal{P}$. Second, a superframe applies changes to a subframe *without* passing through the subframe's admissibility (Graphic Equalisation: admissibility is self-governance, not armour — a graph's grammar governs its self-originated transformation, not world-caused forcing). So the interference is the superframe applying its combined-amplitude structure to the particle; the particle does not select a path by its own rules, and cannot veto the combination.

Corollary 9.2 (The double slit, duality, and delayed choice)**.** *Both paths are admissible continuations in* $\mathcal{P}$*; the superframe applies their combined amplitudes at the screen, giving fringes. A which-path coupling collapses before the screen — one continuation realised, the other no longer in the sum — so the superframe applies a single-path change and the fringes vanish. Wave–particle duality is the map/event distinction: the "wave" is the amplitudes combining (the*

map, the held $\mathcal{P}$), the "particle" the realised outcome (the event, one continuation collapsed), and "which is it really" names no difference. Delayed choice adds no retrocausality: collapse attaches to the actual interaction, not a retroactive rewrite (cost is a local write; no write reaches the past), so interference-or-not depends only on whether the which-path coupling occurs.

Honest boundary: this grounds the shapes — superposition as possibility, interference as phase-combination, the double slit, duality, and delayed choice without retrocausality. The quantitative fringe pattern needs the amplitude phases computed, i.e. the Hilbert formalism, which is the Born-rule frontier of Section 8.

10 The accelerator: forced concurrency and decoherence

The same c bounds concurrency, and this gives the high-energy regime a reading. Within a frame, change is self-change and strictly ordered, so forcing two changes onto one rule-step is the paid exception; a particle accelerator is the machine that pays it, spending energy to push particles' interactions to the same step, where the frame — built to sequence — cannot order them and the interaction is *decoherent* in that frame. The energy divergence as a particle approaches c is the change-rate ceiling: two changes cannot be crowded closer than a step, and forcing them there costs more the closer one gets. The strong reading is "energy is the price of forced concurrency, and forced concurrency is frame decoherence"; anything with a number is weak.

11 Dimensional constants are frame-units; dimensionless ones are not

The reading of c as a frame's own base unit generalises. Each dimensional constant is the base unit of one of a frame's own economies — 1 in-frame, a number only cross-frame.

- c is the change-rate unit (Section 2).
- $\hbar$ is the *aperture quantum*, the smallest resolvable cell: observation is funded by finite internal-change throughput, so fine resolution in one dimension trades against another and the trade floors at a minimum cell — which is the uncertainty principle, $\Delta x\,\Delta p \geq \hbar/2$ being the finite-aperture budget stated as a bound, with $\hbar$ its quantum.
- k_{B} is the information–energy conversion: Landauer's $k_{\mathrm{B}}T\ln 2$ (Landauer, 1961) — the floor cost of change of Graphic Equalisation's economics — is one bit's change converted from the information frame into the energy frame at that rate.
- G is the change-load–geometry coupling, mass deforming the containing frame.

That physics sets $c = \hbar = k_{\mathrm{B}} = G = 1$ (natural units) is then not a convenience but a recognition: these are frame-units, so they read 1 in a frame's own terms, and their familiar magnitudes are cross-frame artefacts — the frame's unit against a human metre, second, kilogram, kelvin. "Why is c that number" is a question about the comparison, not the substrate.

Proposition 11.1 (The dimensional/dimensionless boundary)**.** *This dissolves the* dimensional *constants and touches the* dimensionless *ones not at all. The fine-structure constant $\alpha \approx 1/137$ and the mass ratios cannot be set to* 1 *by any unit choice; they are genuine cross-frame* ratios*, real numbers with no dimension to absorb, and the framework says nothing about their values.*

It sharpens the real question rather than answering it: the numbers that need explaining are the dimensionless ones, while the pseudo-mysteries — why c, why $\hbar$ — are unit conventions.

This is the standard observation that the number of fundamental *dimensional* constants is a matter of convention (Duff, 2002), landing on the side that they are units; the dimensionless remainder is left exactly as open as it was found.

12 The Planck scale, quantization, and fine-tuning

Three consequences follow. First, the *Planck scale* is where the dimensional units meet at 1 simultaneously ($\ell_P = \sqrt{\hbar G/c^3}$ and its companions): the aperture cell ($\hbar$), the change-rate ceiling (c), and the change-load–geometry coupling (G) reach their unit at once. Below it a frame cannot resolve, cannot order, and horizons its own geometry, so nothing there is representable — the " ? " of Graphic Equalisation's epistemic boundary with a location. The Planck scale *is* the frontier, which is why physics predicts its own breakdown there: not a failure of theories but the exhaustion of a frame's units, below which the pre-relational substrate begins.

Second, *quantization* is the aperture being cellular: because $\hbar$ is a minimum cell rather than an infinitesimal, action and energy come in $\hbar$-steps, so the "quantum" of quantum mechanics is Graphic Equalisation's discrete sufficiency read at the resolution axis — the granularity of the aperture, not a separate granularity of the world.

Third, the *dimensionless* constants are contingent rule-content: the framework cannot derive α or the mass ratios, but it can say what kind of thing they are — the contingent specifics of this substrate's rule-set (Graphic Equalisation's nothing-compulsory principle), so "why α" is not "why this necessary value" but "why this rule-set," a question answered with contingency, since a different rule-set gives a different α at no extra machinery. This relocates fine-tuning at its root: the inference "the constants are tuned, therefore design or necessity" assumes they are the kind of thing with a necessary value, whereas the dimensionless ones are units with no value to tune and the dimensionless ones are contingent, so their being tunable is expected rather than remarkable; and the "why this rule-set" is not merely deferred but answered by self-location (Section 15) — we are the observers this rule-set produces.

13 Symmetry and conservation

A symmetry is a transformation that produces no residual: a change the structure does not feel, free and reversible, a no-op in the container's grammar. Noether's theorem, symmetry $\leftrightarrow$ conserved quantity, is then the cost side of that. A quantity is conserved precisely when altering it would cost — produce residual, require a paid write — so the symmetric direction is free and the quantity perpendicular to it cannot move without paying. What a transformation leaves untouched is what it costs to change.

Proposition 13.1 (Momentum conservation is the free-continuation default)**.** *Space-translation symmetry — the rules the same everywhere — produces no residual under translation, and momentum is conserved. But momentum conservation is already the free-continuation default of Graphic Equalisation's momentum: the un-rewritten Narrative, Newton's first law, continuation costing nothing. Noether's momentum case and inertia are one statement — momentum is conserved because only changing it costs.*

Proposition 13.2 (Energy conservation is contingent on time-translation symmetry)**.** *Time-translation symmetry — the rules the same each cycle — conserves the change-budget, which is energy. The failure case is checkable: where the rules effectively change across cycles there is no time-translation symmetry and energy is not globally conserved. This is the standard general-relativistic result that a time-dependent spacetime has no global energy conservation — in an expanding universe photons redshift their energy away — here from the same principle: rules changing across cycles (cosmic expansion, or minting) means no conserved energy.*

Charge follows from gauge symmetry, which is a redundancy of description: a freedom in the representation that changes no physics, in the framework a cross-frame re-description (in the maps, where the continuous lives) producing no in-frame residual, so charge is conserved.

Remark (The gauge group is contingent rule-content). The framework grounds *that* a symmetry yields a conservation law — the shape, residual-free $\leftrightarrow$ cost-to-change — but *which* symmetries hold, the $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$ of the Standard Model, is a fact about this substrate's rule-set, not a necessity. "Why this gauge group" is "why this rule-set," the same relocation as "why α" (Section 12): both are contingent specifics of this universe's grammar, answered with contingency, not derived. The conservation *structure* is universal; the particular symmetries are this universe's.

Honest boundary: this gives why symmetry yields conservation, why momentum conservation is inertia, and why energy conservation fails in cosmology — shapes. It does not derive the Standard Model gauge group, the Noether currents quantitatively, or any number; those are contingent rule-content or ordinary calculations, handed back.

14 The second law

Entropy and its arrow follow from the same economy.

Definition 14.1 (Entropy is what a frame does not distinguish)**.** *The entropy a frame holds is the log-count of the microstructure it carries but cannot tell apart — what falls below its aperture. Entropy is therefore frame- and aperture-relative from the start: a finer aperture resolves more and reads less entropy, a coarser one more.*

This is the Gibbs–Jaynes reading, in which entropy is a property of the observer's coarse-graining rather than of the world in itself (Jaynes, 1957); here it is grounded rather than assumed — entropy is *what this frame does not distinguish.*

Proposition 14.2 (The second law is no-copy accumulated)**.** *Every change discards its predecessor — the prior state goes* NULL *unless paid to keep — so each change pushes structure below the aperture, into the un-distinguished. Change therefore accumulates entropy, because change forgets by default and forgetting is exactly this demotion. The arrow of entropy is the no-copy asymmetry, which is the same asymmetry as accumulated change-cost (the Landauer floors summing) and as causation (a change enables its successor, not its predecessor). Equivalently, entropy is accumulated un-absorbed residual: what no grammar mints stays un-distinguished, so residual piling up is entropy rising.*

Corollary 14.3 (The demon is exorcised, and local order is affordable)**.** *Because k_B converts entropy to energy, Landauer's $k_\mathrm{B}T \ln 2$ is the price of* resolving *one bit — dragging one distinction up out of the un-distinguished, or erasing one back down. Maxwell's demon is exorcised by exactly this: it must pay to distinguish, and that payment is the entropy it thought it removed for free (Bennett, 1982). There is no free distinction, hence no free entropy reduction. Yet local entropy reduction is possible: minting resolves residual and builds structure, but it is a write, so it is paid, and the payment is Landauer heat exported to the surroundings — local order bought by global entropy increase. This is Schrödinger's "life feeds on negentropy" (Schrödinger, 1944) grounded: an organism, a learner, a growing graph is a region that pays to mint local structure and dumps the cost as heat. Life does not violate the second law; it is the second law with a subsidy.*

Remark (The arrows unify). The thermodynamic arrow (entropy up), the causal arrow (change enables forward), and the psychological arrow (memory is present structure of a discarded past) are not three arrows that happen to align; they are one, the no-copy accumulation of change. Every "why do they agree" puzzle in the arrow-of-time literature is asking why three descriptions of a single asymmetry point the same way, and they do because the asymmetry is one: change does not keep its predecessor.

15 Cosmology: the initial state, the seed, and self-location

By Graphic Equalisation's no-start-end result the event order has no intrinsic first, so the beginning of the universe is not an intrinsic edge but a *chosen entry* — a boundary condition imposed from the containing frame. This is why physics supplies initial conditions separately from its laws: the laws are the ordering, which fixes no start, so the entry must come from outside. The Big Bang is that entry, the seed; what is "before" it is not an earlier time but the frontier, the pre-relational substrate, unrepresentable from within, the same " ? " that sits below the Planck scale (Section 12).

Proposition 15.1 (The Past Hypothesis is a seed, hence a hypothesis)**.** *The low-entropy initial state from which the second law takes its direction — the Past Hypothesis — is a seed: the initial rule-and-state configuration installed from the container. Its long-standing status as an assumption rather than a derivation is therefore correct in kind, since a seed cannot be derived from within (a mint cannot authorise its own installation). The arrow of time then splits cleanly: its* mechanism *is no-copy accumulation (Section 14), and its* origin *— that there is a low-entropy state to rise from — is the seed. The direction points away from the imposed low-entropy start; the running is no-copy.*

Proposition 15.2 (Why this seed: self-location)**.** *Why* this *seed — and, equally, why this α, this gauge group — is answered, not declined, and the answer is self-location. A map is referenced from "you are here," and the seed is the here: the origin the frame is drawn from, as $c = 1$ is the unit it is measured from. The observer is not independent of it — by frame-relative identity a different seed is a different observer — so "why this seed rather than another" assumes a trans-seed observer who could compare, and there is none (Axiom -1). It is that way because we are this way, and we are this way because it is that way: one determination read from inside, not two facts one explaining the other.*

This inverts the fine-tuning inference. The constants are not tuned *to* a pre-existing observer; the observer is what those constants produce, so the apparent coincidence is the co-determination of observer and seed mistaken for a tuning aimed at a target. It is the same move as being surprised that the Graph Law applies to everything: it applies because that *is* what everything, measured, is — a constitutive identity, not a contingent breadth, the discriminator being independent re-derivation (Graphic Equalisation's composition falsifier). What remains out of reach is only *derivation*: no formula computes α or the seed. The *why* is answered; the *value* is not derived. Reference points take no "why" — they are where the why is asked.

16 Why 3+1 dimensions, and why a rock has no time

Time is not a must. A frame whose rules are not firing — inert, a rock — has no time at all: nothing happens in it, there is no succession, and in its own frame it merely *is*. What an external clock calls "later" is, for the rock, only another *is*; the weathering it undergoes is the *supergraph's* succession applied to it from above (admissibility being self-governance, not armour), not the rock's own. Time belongs to active, inferring frames — those with a firing closure loop to iterate (Graphic Equalisation's cycle-unit) and the inference to represent the succession. The rock has neither, so it has no time; its rules exist but are inert.

Proposition 16.1 (The observed time is one-dimensional)**.** *Every time we have observed is one-dimensional, and self-change gives a reason it tends that way: change is self-change on one structure, strictly ordered — no two of a frame's own changes are concurrent — so a frame lays down one succession at a time. But this is a tendency, not a theorem. Independent successions are no more prohibited than four spatial dimensions are; what selects the single observed time axis is the same filter that selects three spatial dimensions — decoherence and stability — not a*

necessity. So one-dimensional time is observed and selected, not forced; multi-dimensional time is unobserved, not forbidden. Time itself remains optional, absent from every inert frame.

Space is N-dimensional by contingency: the number of positional coordinates the rules address is rule-content, so "why N" is "why this rule-set," answered by self-location (Section 15). Why $N = 3$ specifically is that self-location filtered by stability — Ehrenfest's result that stable orbits and stable atoms exist only in three spatial dimensions (Ehrenfest, 1917), so observers occur in the three-dimensional rule-sets and we are the observers such a rule-set produces. And the $3 + 1$ *signature*, time set apart from space, is the succession/position distinction: time is the change-axis, the one succession, space the config-axes, the many positions; one time and N space is one-succession-and-many-positions read as a signature.

Honest boundary: time is not necessary (an inert frame has none); the time we observe is 1-dimensional, decoherence-selected rather than forced (multi-dimensional time is unobserved, not prohibited); space is N-dimensional contingently, $N = 3$ by self-location and stability, not by necessity; and the metric signature is the succession/position distinction as a shape, not a derived quantity.

The minimal universe

Everything here is a graph, so a rock and a universe are one kind of thing, differing in scale, not in category (Graphic Equalisation's self-similarity: the runner is graph-that-runs at every scale). A rock *is* a universe — the minimal one — and that makes it an existence proof about universes in general.

Proposition 16.2 (Compulsory versus contingent)**.** *Anything not compulsory for a rock is not compulsory for a universe. The rock does without time (it is inert, no succession), without cycles or closure, without inference or observation, without history or records, without self-originated change, and without a running arrow — so none of these is compulsory for a universe. What even the rock cannot do without is the compulsory core: structure (it* is *nodes and relations), rules (graph data, present though inert), being a subgraph of a supergraph (which applies changes to it from above), and an admissibility grammar (the rules that would gate its self-change, unfired). That core is the whole of what a universe must have.*

This is the general form of "time is optional," and it licenses the verdicts of this paper. Wherever a shape was called contingent rule-content rather than necessity — the number of spatial dimensions, the gauge group, the dimensionless constants, the arrow, the value of c — the rock is the reason: the minimal universe does without it, so it cannot be compulsory. Physics ordinarily takes time, causality, and the second law's arrow as features a universe must have; the rock shows they are features an *active* universe has. And the block universe is not prohibited either: an MRI of a rock is a block-universe progression — the succession held as one static laid-out frame (Graphic Equalisation's history-as-held-frame). What Axiom -1 withholds is only a *completed* block with a fixed future already in it; a block of the succession *so far* is an ordinary held record, open at the frontier, not forbidden.

Honest boundary: the argument is as strong as the self-similarity it rests on (rock and universe as one kind, differing by scale). It does not claim a universe *is* inert like a rock — an active universe has time, change, an arrow. It claims only that a universe *need not* have what the rock does without: those features are contingent on activity, not compulsory to being a universe at all.

Running and recording: Conway, a movie, and the machinima between

Recording spatialises time. A succession held as a record — an MRI of a rock, a film strip, a worldline drawn on paper — lays the temporal order out as a static structure, a +-like axis (the

block's extra dimension). This reads the metric signature: the − axis is the one being *lived*, the frame's own succession run by its cycles, open at the frontier; the + axes are *held* — spatial positions, and any recorded time. The sign tracks lived-versus-held, not a property intrinsic to the axis; the block feels dead because a recorded succession is not run.

The extremes of the run-versus-record axis are Conway and a movie. Conway's Game of Life is pure running: a rule and a seed, everything computed live, nothing recorded, and no way to know the outcome but to run it (computational irreducibility). A movie is pure record: every frame laid out, played back, nothing computed — the block universe, known by reading rather than running. Between them is scripted machinima — a running engine following a predetermined script, part computed live, part laid down in advance — and that is where real frames sit: rules fire (Conway-like) while history is held (movie-like). It is the same axis as generation-versus-record: store the generator and run it (Conway, small but irreducible) or store the frames and read them (a movie, large but immediate), with the world's own compressibility deciding which is cheaper.

Proposition 16.3 (The past can be a movie; the future is Conway)**.** *A block is only ever of the past. What has been run can be recorded and laid out — a movie of the succession so far, readable without re-running. What has not been run cannot be recorded, because there is nothing yet to record: the future is determined by the rules but* uncomputed*, Conway-style, so it is not laid out and cannot be read ahead — you must run it. The open frontier of Axiom* −1 *is therefore not undetermined-ness but unrun-ness: the future is open because it is uncomputed, not because it is unfixed, and you cannot record what has not run.*

Honest boundary: this applies Graphic Equalisation's generation-versus-record axis to whole universes, with Conway and film as its two poles (Conway is itself a seed in the substrate). It adds no formal machinery — it identifies the lived/recorded and generation/record axes as one, and locates the open future as the unrun edge of the run, not as indeterminacy.

Corollary 16.4 (Free will is the in-frame run; a story demonstrates it)**.** *Free will is a frame running its own decision. In a frame's own run the decision is live and unrun-to-it — the frame must run its deliberation to reach the outcome — and that running* is *the choosing: the outcome is fixed by the rules yet knowable only by running, so the frame experiences computing it as choosing it. This needs no indeterminacy, and a story is the perfect demonstration. A book or a movie is a fully recorded succession — every word or frame fixed, zero randomness, readable ahead by the reader — and yet its characters have complete free will: they deliberate, choose, and are responsible, in frame. Both hold at once because they are two frames. The character's freedom is its in-frame run; the reader's foreknowledge is a superframe holding the finished record (a movie), and holding the record never reaches into the character's run to remove the freedom. So foreknowledge is not unfreedom — not even complete foreknowledge, the reader who knows the ending or the author who wrote it — because knowing the recorded run and being the live run are different frames. Randomness would only subtract: a random run is not more free, it is less the character's own. And responsibility survives for the reason the freedom does — the run is the character's own (self-change is the only change; the rules running are the character) — so a determined, recorded, even foreknown act is still that character's act.*

Honest boundary: this is a compatibilist stance — free will as the frame's own in-frame run, frame-relative and robust to a superframe's complete foreknowledge, with no appeal to indeterminacy. It claims the shape of the free/determined dissolution, not a decision procedure, and takes no position on which acts a given reader can in fact foreknow (that is the run/record split of Proposition 16.3).

Remark (Author, character, reader: unrun-ness is absolute)*.* A story stacks three frames on one succession. The author *runs* it — writing is Conway, generated live, and the author must write to know the ending (authorship is computationally irreducible; the commonplace that "the

characters took on lives of their own" is the author unable to read ahead their own creation). The character *lives* it — the in-frame run of Corollary 16.4. The reader *replays* it — the finished record, a movie. One succession, three frame-readings: the run/record axis is a fact about frame, not about the system. And the author settles the objection to Proposition 16.3 that a high enough superframe might read the future ahead — even the creator gets no shortcut. The author is Conway to their own book while writing it; the movie exists only *after* the run, for everyone including the author. Foreknowledge is always post-run record-holding, never pre-run read-ahead. Unrun-ness is absolute: no frame, creator included, reads an unrun future; every frame either runs it or replays an already-run record.

Remark (There is no non-actual frame; imagination is real). Imagination is not the running of a non-actual frame — there is no such thing. Every frame that runs is real graph data really running, so actuality is not a property some frames have and others lack; every frame is actual. An imagined dragon, a planned move, a character in a book is a real frame, really run (real computation), not a pretend one. The only difference between an imagined frame and a perceived one is frame-relative grounding: what the frame is anchored to, relative to another frame. A perception is grounded in the observation frame (sensory-anchored); an imagination is grounded internally, in generation or in a held or communicated record. Both are real; they differ in what they are grounded in, not in whether they are. So "is it real?" is malformed as a question about actuality — everything that runs is — and the well-formed question is "what is it grounded in?": observation, memory, generation, or a communicated record. A hallucination is then a real frame mistagged as observation-grounded (a grounding error, not a false frame); a shared fiction is a real frame grounded in a communicated record (the book of Remark 16); a plan is a real frame grounded in projection. This is why imagination does work — planning, design, and counterfactual reasoning are real frames really run, not pretence — and why fiction moves: the character's frame runs, really, in the reader. Death, in the same terms, is not a frame going non-actual (nothing does) but a run that stops, leaving frames grounded only in a record and re-runnable by others; reading and remembering are that re-running — a record-grounded frame run again into a live grounding.

Honest boundary: actuality is not a frame-property (every running frame is actual); the real distinction is frame-relative grounding, and this remark does not claim every frame is equally grounded in observation — the dragon is not — only that every frame is equally real, differing in what it is grounded in. It claims the shape (reality is running; the distinction is grounding), not a metric of grounding.

Remark (Death is when the self's rules cease to be re-runnable in the same coherent self). Death is, critically, the rules within the self ceasing to be re-runnable in the same coherent self entity. Not every stop is death: a pause — sleep, anaesthesia, a recoverable coma — is a stop whose rules remain re-runnable in the same coherent self, and the self resumes as itself. Death is the stop that does not resume: the coherence that made these rules one self is gone, so they cannot be re-run as that self, even if the graph-data they were persists scattered as a record. This is the identity criterion — the self persists exactly while its rules stay re-runnable as the same coherent entity — and it is why a pause is survivable and death is not; the self-graph goes inert for good, a record where it was a run (Section 16's rock, now grounded only in what remains of it).

What ends is only the self's own run. The supergraph runs on, and two things the self did persist in it. Its changes persist: rule-actions are pointers, work done once, so every write the agent made to the supergraph is the supergraph's now, permanent, and death does not undo it. And its copies persist: every frame the agent communicated, taught, wrote, or built runs on in other frames, grounded there in the record it left (the reader re-running the author of Remark 16). But those copies are not the self resuming — they run as fragments in other coherent selves, not as this coherent self re-run — so their persistence is not the self's continuation. The self can no longer self-change, and self-change was the only change it had; the changes it already made and the copies already running elsewhere are no longer its to stop. They keep running without it.

Honest boundary: this is the run/record, pointer, and copy machinery applied to a self, and it describes what death *is* in the substrate — the self's rules ceasing to be re-runnable as the same coherent self, while its writes and its copies persist — not any continuation of the self's own run. There is no such continuation: nothing re-runs as the same coherent self, and what persists (consequences and copies) is not that self running. The identity criterion is offered as the discriminator between a pause and death, not as an operational test of when re-runnability has in fact been lost; and this claims the shape, not an account of the subjective experience of dying.

Corollary 16.5 (Personal identity is continuity of the coherent run)**.** *The identity criterion resolves the standard personal-identity puzzles, because the invariant it names is continuity of the coherent run, not preservation of parts or data. Gradual replacement (cell turnover, the Ship of Theseus) is survival: coherence is maintained at every step, so the same coherent self stays re-runnable throughout — the self is the coherent running, not the substrate. A teleporter or an upload is survival exactly when the coherence is carried unbroken; a copy-then-destroy is not, because the copy is a new coherent self grounded in the record while the original's rules ceased to be re-runnable as the same coherent self. Fission — one coherent self becoming two — branches: there are then two coherent continuations and neither is uniquely the self. A backup restored is a new coherent self grounded in the record, not the original re-run, which is why a backup is not immortality (the record-to-run direction starts a fresh coherence unless coherence was carried unbroken). A record — a scan, a backup, a corpse — is never the self; re-running it starts a new coherent self.*

Remark (Death is relative to the coherency of the self)*.* It follows that death is relative to the coherency of the self. The criterion applies within a coherent self, but which coherence counts as "the same self" is a frame-relative choice, so death is indexed to that choice, not absolute. Drawn narrowly — this instant's exact coherent state — the self dies each moment, replaced by the next; drawn broadly — the coherent person tolerant of change across a life — it dies once, at the irreversible break. A deep transformation ("I am not the person I was") is a death relative to the narrow coherence and a survival relative to the broad one, both true at once. There is no absolute self, so there is no absolute death — only a death per coherence-boundary. This makes death relative, not unreal: each boundary's death is real for that boundary — the broad life-self's coherence does break irreversibly, and that death is as real as any — so relativity means many real deaths at many boundaries, never none.

Honest boundary: identity is continuity of the coherent run (a shape, not a sharp threshold for how gradual a change counts as unbroken), and death is relative to a chosen coherence-boundary (many real deaths, none absolute) — which relativises *when* death occurs and to which self, without denying *that* it occurs at each boundary.

17 Identical particles are functional identity

Graphic Equalisation's Axiom −1 holds that two things with no difference that makes a difference are the same, and its no-frame-free-identity theorem that one cannot say which of two is which independently of an individuating frame. This is the quantum treatment of identical particles exactly: two electrons carry no label, and "which is which" names no difference and so has no content. Classical physics labels identical objects — treats two identical bodies as distinct in principle — and that label is a fiction, a distinction imposed where none exists.

Proposition 17.1 (Quantum statistics, and the Gibbs paradox, are functional identity)**.** *Classical Maxwell–Boltzmann counting labels identical particles and over-counts arrangements; quantum Bose–Einstein and Fermi–Dirac counting treats them as one where they are indistinguishable. The Gibbs paradox — the spurious entropy of mixing two identical gases — resolves by not*

labelling them, which is Axiom −1, and ties to Section 14: entropy is what a frame distinguishes, so identical particles, having nothing to distinguish, add no mixing entropy. The $1/N!$ *Gibbs correction is functional identity made quantitative.*

Corollary 17.2 (Pauli exclusion is "distinct things need a distinction")**.** *If two fermions were in the same state — the same in every quantum number — they would carry no distinguishing distinction, and by Axiom −1 would be one, not two. To be two they must differ somewhere. The exclusion principle is therefore the requirement that distinct things be distinguishable: there is no such thing as two indistinguishables in one state, because that "two" names no difference. Exclusion is the arithmetic of identity, not a force.*

Remark (The honest limit: spin–statistics)*.* Indistinguishability and fermion exclusion are clean here; the boson side is not. Why bosons pile into one state as a count while fermions cap at one, and the minus sign under exchange tied to a 2π rotation, is the spin–statistics theorem, which needs structure this account does not supply. The framework grounds *that* particles are indistinguishable and *why* fermions exclude, and hands the boson/sign/spin–statistics content back untouched — the same shape/number division as elsewhere.

18 Positioning and prior art

The pieces are individually established; the contribution is the conjunction under one change-economy and the mechanisation of shapes prior art postulates, geometrises, or offers heuristically.

Deriving c, and mass as internal motion. That c should be *derived* rather than postulated has precedent by a different route. Ignatowsky (Ignatowsky, 1911) and, cleanly, Lévy-Leblond (Lévy-Leblond, 1976) obtain the Lorentz transformation from the relativity principle together with homogeneity, isotropy, and group structure alone, with an invariant speed *emerging* rather than assumed and its value left empirical; those derivations use symmetry as the engine where the present one uses the change-substrate — c as each frame's own change-throughput unit, 1 by construction. The reading of mass as the budget reserved for internal change has two standing precedents, offered heuristically rather than mechanised: Schrödinger's *zitterbewegung*, the electron's rest energy as internal motion at c (Schrödinger, 1930), developed as a rest-mass-from-internal- circulation interpretation by Hestenes (Hestenes, 1990); and the elementary result that confined massless radiation — light in a box — carries rest mass E/c^2, internalised c-motion appearing as inertia. The budget-split itself is the constant-magnitude four-velocity of Minkowski geometry (Minkowski, 1909).

Relativity as postulate. Einstein's special (Einstein, 1905) and general (Einstein, 1916) relativity take c-invariance and the equivalence principle as postulates; the present account renders the first a theorem (Proposition 2.1) and the second a common cause (Section 4), at the cost of claiming only the shapes.

Entropy and the demon. The observer-relative reading of entropy is Jaynes's (Jaynes, 1957); the resolution of Maxwell's demon by the cost of information is Landauer's and Bennett's (Landauer, 1961; Bennett, 1982); life as locally paid negentropy is Schrödinger's (Schrödinger, 1944). The present account ties all three to one substrate law — cost is the write.

Dimensional constants. That the count of fundamental *dimensional* constants is conventional is Duff's position in the Duff–Okun–Veneziano exchange (Duff, 2002); the present account gives the reason (they are frame-units) and draws the boundary at the dimensionless remainder.

19 Falsifiability commitments

Each is exposed to attack; the scope caveat of Section 1 applies throughout.

1. c **as frame-unit.** Fails if a frame can measure its own base change-rate as anything other than 1, or if c-invariance requires a postulate that cannot be recovered as the unit reading.

2. **One budget, two dilations.** Fails if velocity time dilation and gravitational time dilation cannot be exhibited as the same fixed budget spent on motion and on change-load respectively — if the equivalence principle is not their common cause here.

3. **Dimensional/dimensionless boundary.** Fails if a genuinely dimensionless constant can be set to 1 by unit choice, or if a dimensional constant carries an irreducible value not absorbable into a frame's units.

4. **Second law as no-copy.** Fails if entropy can rise in a frame whose changes preserve their predecessors, or fall in one without a paid write exporting cost.

5. **Indistinguishability is functional identity.** Fails if identical particles carry a frame-free label distinguishing them, or if the Gibbs $1/N!$ correction and Fermi exclusion cannot be read as "distinct things require a distinguishing distinction." (The boson/spin–statistics content is not claimed.)

6. **Measurement is frame-relative collapse.** Fails if a frame-independent fact about whether collapse has occurred is required to describe a Wigner's-friend scenario — if the friend's outcome and the outside frame's superposition cannot both stand as the event and the map of two frames. (The Born-rule probabilities are not claimed.)

7. **Entanglement is a superframe dimension.** Fails if the correlation of an entangled pair is a subframe-local quantity rather than a joint dimension NULL in each subframe, or if a reduced single-particle state can be pure — i.e. if the mixedness is not the inaccessibility of a superframe dimension. (The Tsirelson bound and correlation strengths are not claimed.)

8. **Conservation is the cost side of symmetry.** Fails if a continuous symmetry (a residual-free transformation) carries no conserved quantity, or if a conserved quantity persists across a genuine change in the rules — in particular, if global energy is conserved in a time-dependent (rules-changing) regime such as cosmic expansion. (The gauge group is claimed to be contingent, not derived.)

9. **The initial state is a seed, answered by self-location.** Fails if "why this seed / constant / gauge group" can be posed from a trans-frame standpoint that could have obtained a different one — if there is an observer-independent "we" separable from the seed. (The claim is that the *why* is self-location and the *value* is not derived; both together, not either alone.)

10. **Amplitude is directional Narrative.** Fails if the complex amplitude cannot be identified with a magnitude-times-direction operative-history structure, or if the Born square admits no closed-loop (conjugate) reading. (The square is offered as a candidate via closure; its rigorous tie to the Hilbert inner product and Gleason's theorem is stated as not yet done, not as achieved.)

11. **Interference is possibility plus superframe forcing.** Fails if interference requires the particle's own admissibility to select among paths, or if a which-path coupling that collapses before the screen does not remove the fringes — i.e. if superposition is not the possibility structure and the superframe does not apply the combination without passing through the subframe's grammar.

12. **Time is optional; the observed time is one-dimensional.** Fails if an inert frame (no firing rules) nonetheless has an in-frame succession. (That the observed time is 1-D is claimed decoherence-selected, not forced — multi-dimensional time is unobserved, not prohibited; that space is N-D with $N = 3$ is likewise contingent and self-located; and the block universe is a held record, not forbidden.)

13. **The future is unrun, not unfixed.** Fails if a determined future can be read ahead without running it — a shortcut past computational irreducibility for a genuinely irreducible rule — or if a block can contain a not-yet-run future. The past-as-movie / future-as-Conway split predicts records only ever trail the run, never lead it.

14. **Free will is the in-frame run, not indeterminacy.** Fails if a fully recorded, zero-randomness system (a story) cannot exhibit the form of free will — deliberation, choice, responsibility — in its characters' frames, or if a superframe's complete foreknowledge demonstrably removes a subframe's in-frame freedom. The claim predicts foreknowledge and freedom coexist across frames.

15. **Actuality is running, not a frame-property.** Fails if some running frame is demonstrably non-actual (real graph data that runs yet is not real), or if the imagined/perceived distinction requires an actuality property rather than frame-relative grounding. The claim predicts imagination does real computational work (planning, design) as real runs, not pretence.

16. **Death is loss of re-runnability in the same coherent self.** Fails if a pause that preserves re-runnability as the same coherent self (sleep, a recoverable coma) counts as death, or a stop that loses it counts as survival — the criterion is re-runnability in the same coherent entity, not the mere stopping. Also fails if a self's writes to the supergraph vanish when its run stops (work-done-once violated), or if frames it copied into other frames cease when it does. The claim predicts consequences and copies persist, that copies are not the self resuming, and that nothing re-runs as the same coherent self.

17. **Identity is continuity of coherence; death is relative to the coherence-boundary.** Fails if two systems agreeing on all parts and data but differing in coherence-continuity share identity, or if continuity of coherence through total part-replacement destroys it (gradual replacement must survive, copy-then-destroy must not, fission must branch). And fails if there is a single absolute self whose end is the one death independent of any coherence-choice, or if no coherence-boundary's death is real — the claim predicts many real deaths at many boundaries, none absolute.

18. **Scope.** The paper claims shapes only; it is falsified as *overclaiming* if any passage is read to derive a field equation, a dimensionless constant, or a quantum-gravity result — none is asserted.

References

John S. Bell. On the Einstein Podolsky Rosen paradox. *Physics Physique Fizika*, 1:195–200, 1964.

Charles H. Bennett. The thermodynamics of computation — a review. *International Journal of Theoretical Physics*, 21(12):905–940, 1982.

Michael J. Duff. Comment on time-variation of fundamental constants. arXiv:hep-th/0208093, 2002. (Duff–Okun–Veneziano, "Trialogue on the number of fundamental constants.")

Paul Ehrenfest. In what way does it become manifest in the fundamental laws of physics that space has three dimensions? *Proceedings of the Amsterdam Academy*, 20:200–209, 1917.

Albert Einstein. Zur Elektrodynamik bewegter Körper. *Annalen der Physik*, 322(10):891–921, 1905.

Albert Einstein. Die Grundlage der allgemeinen Relativitätstheorie. *Annalen der Physik*, 354(7):769–822, 1916.

Andrew M. Gleason. Measures on the closed subspaces of a Hilbert space. *Journal of Mathematics and Mechanics*, 6:885–893, 1957.

David Hestenes. The zitterbewegung interpretation of quantum mechanics. *Foundations of Physics*, 20(10):1213–1232, 1990.

Wladimir A. von Ignatowsky. Einige allgemeine Bemerkungen zum Relativitätsprinzip. *Physikalische Zeitschrift*, 11:972–976, 1911.

Edwin T. Jaynes. Information theory and statistical mechanics. *Physical Review*, 106:620–630, 1957.

Simon Kochen and Ernst P. Specker. The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics*, 17:59–87, 1967.

Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961.

Jean-Marc Lévy-Leblond. One more derivation of the Lorentz transformation. *American Journal of Physics*, 44(3):271–277, 1976.

Hermann Minkowski. Raum und Zeit. *Physikalische Zeitschrift*, 10:75–88, 1909. Address delivered 1908.

Carlo Rovelli. Relational quantum mechanics. *International Journal of Theoretical Physics*, 35:1637–1678, 1996.

Erwin Schrödinger. Über die kräftefreie Bewegung in der relativistischen Quantenmechanik. *Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-mathematische Klasse*, 24:418–428, 1930.

Erwin Schrödinger. *What Is Life? The Physical Aspect of the Living Cell*. Cambridge University Press, Cambridge, 1944.